

KNEO AGENT CLIENT

API Reference

Typed Python SDK + adapter toolkit for the Kneo Agent Platform /v1 .

Single-document reference for every public class, dataclass, exception, and helper that `kneo-client` exposes. Sibling to the `kneo_agent` SDK reference; same dark-themed layout, same hand-maintained discipline.

VERSION

v0.5.0

PINNED TO KNEO_SERV

v0.5.0 (/v1 only)

PYTHON

≥ 3.12

The user-facing prose ships separately as the Kneo Agent Client User Guide PDF. Source markdown lives under `docs/user/` on the repo. Architecture lives in `docs/dev/architecture.md` and the design handout PDF.

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Every public class and helper, organized for both screen reading and printing. Adapted from the `kneo_agent` SDK reference's dark-themed layout; rendered to PDF via the same Puppeteer pipeline.

v0.5.0

Python ≥ 3.12

kneo_serv /v1 · 0.4.x

What's new in v0.5.0

A small, focused cut that closes the `audit.list` deprecation window, refreshes the pinned `kneo_serv` spec to `v0.5.0`, and adds a typed error for the platform's new backpressure signal. The pin moves but the `/v1` contract is byte-identical to `v0.4.0` — only `info.version` changes — so the generated client is unchanged.

Headline pieces:

- **`audit.list` dead kwargs removed (breaking).** `principal`, `offset`, `sort_by`, and `sort_order` — deprecated in 0.4.0, never honored by `/v1/audit-events` (which accepts only `event_type` / `run_id` / `limit`) — are now **removed** from the signature. Passing any is a `TypeError`. Migration: drop them from your call (the platform never used them, so there is no behavior change beyond the call no longer accepting the argument).
- **Spec pin refreshed to `kneo_serv v0.5.0`.** `v0.5.0` is an operability minor (worker-pool concurrency, a root `/metrics` endpoint, dead-lettering); its `/v1` surface is identical to `v0.4.0`, so no wrappers changed. The compatibility matrix gains the `0.5.0` row.
- **New `KneoServiceUnavailable (HTTP 503)`.** `kneo_serv v0.5.0` sheds load on `POST /v1/runs` with `503 + Retry-After` once its run queue is saturated. The transport already auto-retries 503 (honoring `Retry-After`); once retries are exhausted the caller now gets a typed `KneoServiceUnavailable` (a `KneoServerError` subclass) carrying `.retry_after` for longer caller-side back-off.

See the [0.5.0 release notes](#) for the full summary including the breaking-change upgrade note, the spec-pin refresh, and the new error type. The [0.4.0 release notes](#) cover the prior cut's `audit.list` deprecation and smoke-template fix.

Introduction

`kneo-client` is the shared client layer behind **Kneo Agent Dashboard** (operations) and **Kneo Agent Studio** (development). It owns the operational semantics of talking to a Kneo Agent Platform instance over its `/v1` HTTP API so that downstream products do not each re-invent them.

Three layers, strict dependency direction *generated* → *core* → *adapters*:

- `kneo_client._generated` — *private*. Generated from the pinned `kneo_serv` OpenAPI spec via `openapi-python-client`. External code must not import from this subpackage.
- `kneo_client.core` — handwritten cross-cutting layer: `Transport`, `Profile`, `ApiKeyAuth`, `RetryPolicy`, `idempotency`, `pagination`, `request IDs`, `redacted logging`, `normalized errors`.
- `kneo_client.platform` and `kneo_client.agent` — domain-shaped adapters built on `Transport`. `PlatformClient` for the Dashboard, `AgentClient` for the Studio.

Public entry point is a unified `KneoClient` with `.platform` and `.agent` namespaces backed by a single shared `Transport`.

Privacy boundary. Anything under `kneo_client._generated` is private implementation. Its surface can change between any two `kneo-client` releases without notice. If a generated type or helper is useful to external consumers, it gets re-exported through `core`, `platform`, or `agent`; otherwise it stays internal.

For the architectural rationale see `docs/dev/architecture.md` on the repo. For each major design choice see the ADRs under `docs/dev/adrs/`.

KneoClient

The public entry point. Owns one [Transport](#) and mounts `.platform` and `.agent` namespaces backed by it. Async context manager; closes its transport on exit.

class KneoClient CLASS

Async client for a Kneo Agent Platform instance.

Attributes

ATTRIBUTE	TYPE	DESCRIPTION
-----------	------	-------------

<code>platform</code>	PlatformClient	Operational surface — runs, traces, audit, credentials, policies, health.
<code>agent</code>	AgentClient	Development surface — spec validate / compile / explain / policy_report / run.
<code>profile</code>	Profile	The resolved profile this client is bound to.

```
def __init__(profile: Profile, *, retry_policy: RetryPolicy | None = None) → None
```

Build a `KneoClient` bound to `profile`. The transport is constructed eagerly; nothing happens on the network until the first `.platform.*` or `.agent.*` call.

PARAMETER	TYPE	DEFAULT	DESCRIPTION
<code>profile</code>	Profile	—	Resolved profile carrying URL, API key, scheme, and timeout.
<code>retry_policy</code>	RetryPolicy <code>None</code>	<code>None</code>	Override the default retry policy (3 attempts, exp backoff). See RetryPolicy .

```
@classmethod def from_profile(name: str | None = None, **overrides: Any) → KneoClient
```

Resolve a profile via [load_profile\(\)](#) and build a `KneoClient`. Convenience for the common case of "use the standard config-file / env / kwargs resolution chain".

PARAMETER	TYPE	DEFAULT	DESCRIPTION
<code>name</code>	<code>str</code> <code>None</code>	<code>None</code>	Profile name to load. Falls back to <code>\$KNEO_PROFILE</code> then "default".

**overrides	Any	—	Forwarded to load_profile : config_file , url , api_key , auth_scheme , timeout .
--------------------	-----	---	--

```
async def aclose() → None
```

Close the underlying transport. Idempotent.

```
async def __aenter__() → KneoClient · async def __aexit__(*exc) → None
```

Async context manager. Use `async with KneoClient.from_profile() as client:` to ensure the transport closes on exit.

Example

```
import asyncio
from kneo_client import KneoClient

async def main():
    async with KneoClient.from_profile() as client:
        ready = await client.platform.health.readyz()
        print(f"ok={ready.ok}")

        run = await client.platform.runs.create({"spec_id": "my-spec"})
        terminal = await client.platform.runs.wait_for_completion(run.run_id,
            timeout=120)
        print(f"final={terminal.status}")

    asyncio.run(main())
```

Transport

The request engine. `Transport` is the only layer that does I/O; every adapter call eventually routes through its `request()` method.

Owens: the `httpx.AsyncClient`, the auth flow, the retry loop, idempotency-key injection, request-ID injection, redacted logging, and the error-to-exception mapping point.

class Transport CLASS

Async HTTP transport for a Kneo Agent Platform instance.

Attributes

ATTRIBUTE	TYPE	DESCRIPTION
<code>profile</code>	<code>Profile</code>	The profile this transport is bound to (read-only).

```
def __init__(profile: Profile, *, http_client: httpx.AsyncClient | None = None, retry_policy: RetryPolicy | None = None) → None
```

PARAMETER	TYPE	DEFAULT	DESCRIPTION
<code>profile</code>	<code>Profile</code>	—	Resolved profile.
<code>http_client</code>	<code>httpx.AsyncClient None</code>	<code>None</code>	Optional pre-built client. When supplied, the caller owns its lifecycle; <code>aclose()</code> will not close it. Base URL and auth are <i>not</i> overridden on a passed-in client.
<code>retry_policy</code>	<code>RetryPolicy None</code>	<code>None</code>	Defaults to <code>RetryPolicy()</code> (3 attempts, exp backoff with jitter).

```
async def request(method: str, path: str, *, json: Any = None, params: Mapping | None = None, headers: Mapping | None = None, idempotency_key: str | None = None, request_id: str | None = None) → httpx.Response
```


Send an HTTP request with auth, retries, idempotency, and error mapping.

PARAMETER	TYPE	DEFAULT	DESCRIPTION
<code>method</code>	<code>str</code>	—	HTTP method (GET , POST , ...).
<code>path</code>	<code>str</code>	—	Path relative to the profile's base URL (e.g. /v1/runs).
<code>json</code>	<code>Any</code>	<code>None</code>	JSON body.
<code>params</code>	<code>Mapping[str, Any] None</code>	<code>None</code>	Query string parameters.
<code>headers</code>	<code>Mapping[str, str] None</code>	<code>None</code>	Extra headers. Auth, Idempotency-Key , and X-Request-ID are added automatically and override anything in this mapping.
<code>idempotency_key</code>	<code>str None</code>	<code>None</code>	Override the auto-generated POST idempotency key. Validated against the 256-character platform limit. Ignored on non-POST methods.
<code>request_id</code>	<code>str None</code>	<code>None</code>	Override the auto-generated request ID.

RETURNS

`httpx.Response` — the successful response (status < 400).

RAISES

A [KneoError](#) subclass after retries are exhausted. See [Exceptions](#) for the mapping.

Retry behavior

The transport retries on:

- Transport-level errors from `httpx` (DNS, connect, read timeout, TLS).
- HTTP status in `RETRYABLE_STATUS_CODES = {429, 502, 503, 504}`.

Retries fire only for:

- **Idempotent verbs** — `GET`, `HEAD`, `OPTIONS`, `PUT`, `DELETE`.
- **POST with an idempotency key** — which is always, since the transport auto-injects a UUID4 key on every `POST`.

A `Retry-After` response header on 429 overrides the computed delay verbatim (jitter is not applied on top).

```
async def aclose() → None
```

Close the underlying `httpx.AsyncClient` if this `Transport` owns it. When a caller-supplied `http_client` was passed to `__init__`, the caller's client is not closed.

```
async def __aenter__() · async def __aexit__(*exc)
```

Async context manager. `async with Transport(profile) as t:` closes the transport on exit.

Example: direct use (no adapter)

```
import asyncio
from kneo_client.core.profiles import load_profile
from kneo_client.core.transport import Transport

async def main():
    profile = load_profile()
    async with Transport(profile) as t:
        resp = await t.request("GET", "/v1/healthz")
        print(resp.json())

asyncio.run(main())
```

SyncTransport

Synchronous facade around `Transport`. Runs the async transport in a dedicated background thread + event loop via `anyio.from_thread.start_blocking_portal()`. Each call dispatches into that loop and

blocks until the coroutine returns.

For callers that cannot run an event loop (scripts, notebooks, sync frameworks). The async surface remains the recommended path.

class SyncTransport CLASS

```
def __init__(profile: Profile, *, retry_policy: RetryPolicy | None = None) → None
```

```
def request(method: str, path: str, *, json=None, params=None, headers=None, idempotency_key=None, request_id=None) → httpx.Response
```

Synchronous version of `Transport.request`. Same arguments, same semantics; blocks until done.

```
def close() → None · def __enter__() · def __exit__(*exc)
```

Lifecycle. Use `with SyncTransport(profile) as t:` to close the background portal on exit.

Caveat. `SyncTransport` does **not** mount `.platform` / `.agent` adapters. Those are async-only. Sync consumers wanting wrapped endpoints either glue async-to-sync at the call site, or drop to `t.request(method, path, ...)` directly.

Example

```
from kneo_client.core.profiles import load_profile
from kneo_client.core.transport import SyncTransport

with SyncTransport(load_profile()) as t:
    resp = t.request("GET", "/v1/healthz")
    print(resp.json())
```

Profile & profiles

A *profile* is a frozen dataclass bundling (name, url, api_key, auth_scheme, timeout) — everything Transport needs to talk to one Kneo Agent Platform instance.

class Profile DATACLASS

frozen=True . Pass directly to `KneoClient(profile)` or build via `load_profile()` .

FIELD	TYPE	DEFAULT	DESCRIPTION
name	str	—	Profile name. Informational; used in log records.
url	str	—	Base URL of the platform instance.
api_key	str	—	API key to authenticate with. Treat like any other secret; the redaction-aware logger masks it in header output.
auth_scheme	AuthScheme	BEARER	How to present the key (Authorization: Bearer vs X-Kneo-Api-Key).
timeout	float	30.0	Per-request timeout in seconds.

class ProfileError EXCEPTION

Raised when a profile cannot be resolved (missing url / api_key , malformed TOML, bad scheme, non-numeric timeout, etc.).

load_profile() FUNCTION

```
def load_profile(name: str | None = None, *, config_file: Path | None = None,
url: str | None = None, api_key: str | None = None, auth_scheme: AuthScheme |
```

```
str | None = None, timeout: float | None = None) → Profile
```

Resolve a `Profile` from TOML config + env vars + explicit kwargs. Resolution order (later overrides earlier):

1. TOML at `config_file` (default: `~/.config/kneo/client.toml` via `platformdirs`).
2. Environment variables: `KNEO_URL`, `KNEO_API_KEY`, `KNEO_AUTH_SCHEME`, `KNEO_TIMEOUT`.
`KNEO_PROFILE` selects which TOML section to load.
3. Explicit keyword arguments to this function.

PARAMETER	TYPE	DEFAULT	DESCRIPTION
<code>name</code>	<code>str None</code>	<code>None</code>	Profile name. Falls back to <code>\$KNEO_PROFILE</code> , then <code>"default"</code> .
<code>config_file</code>	<code>Path None</code>	<code>None</code>	Override the default TOML location. Non-existent files are skipped silently.
<code>url</code> , <code>api_key</code> , <code>auth_scheme</code> , <code>timeout</code>	<code>str / AuthScheme / float</code>	<code>None</code>	Override the resolved field. <code>auth_scheme</code> accepts either an enum or its string value.

RETURNS

`Profile` — fully resolved.

RAISES

`ProfileError` — if `url` or `api_key` cannot be resolved from any source, or if the config file is malformed.

```
def default_config_path() → Path
```

Return the XDG-style default location of `client.toml` (`~/.config/kneo/client.toml` on Linux, `~/Library/Application Support/kneo/client.toml` on macOS, etc., via `platformdirs.user_config_dir("kneo")`).

Environment-variable constants

CONSTANT	VALUE	PURPOSE
----------	-------	---------

DEFAULT_PROFILE_NAME	"default"	Profile name when none is specified.
DEFAULT_TIMEOUT_SECONDS	30.0	Default per-request timeout.
ENV_PROFILE	"KNEO_PROFILE"	Selects which TOML section to load.
ENV_URL	"KNEO_URL"	Override profile url .
ENV_API_KEY	"KNEO_API_KEY"	Override profile api_key .
ENV_AUTH_SCHEME	"KNEO_AUTH_SCHEME"	Override profile auth_scheme .
ENV_TIMEOUT	"KNEO_TIMEOUT"	Override profile timeout .

Example: TOML config

```
# ~/.config/kneo/client.toml
[default]
url = "https://kneo.example.com"
api_key = "prod-key"
auth_scheme = "bearer"
timeout = 30.0

[staging]
url = "https://staging-kneo.example.com"
api_key = "staging-key"
```

Example: resolve a profile

```
from kneo_client.core.profiles import load_profile

p = load_profile()                # 'default' from TOML + env
p = load_profile("staging")       # explicit profile
p = load_profile(url="https://ad-hoc", api_key=tok) # explicit kwargs win
```

ApiKeyAuth & AuthScheme

The platform accepts the API key as either `Authorization: Bearer <key>` or `X-Kneo-API-Key: <key>`. `ApiKeyAuth` is an `httpx.Auth` subclass that injects the chosen header on every outgoing request; `AuthScheme` is the enum that picks which.

class AuthScheme ENUM

String-valued enum.

MEMBER	VALUE	HEADER INJECTED
<code>BEARER</code>	<code>"bearer"</code>	<code>Authorization: Bearer <key></code>
<code>KNEO_API_KEY</code>	<code>"kneo_api_key"</code>	<code>X-Kneo-API-Key: <key></code>

class ApiKeyAuth CLASS

Inherits from `httpx.Auth`. Compatible with both `httpx.Client` and `httpx.AsyncClient` because `httpx`'s auth flow is a synchronous generator.

```
def __init__(api_key: str, scheme: AuthScheme = AuthScheme.BEARER) → None
```

PARAMETER	TYPE	DEFAULT	DESCRIPTION
<code>api_key</code>	<code>str</code>	—	Non-empty API key. Empty values raise <code>ValueError</code> .
<code>scheme</code>	<code>AuthScheme</code>	<code>BEARER</code>	Header scheme.

```
def auth_flow(request: httpx.Request) → Generator[httpx.Request, httpx.Response, None]
```

Injects the API key header and yields the request unchanged. Called by `httpx` for every redirect / retry attempt at the transport-level layer.

RetryPolicy

A frozen dataclass describing when and how long to wait between attempts. `Transport` applies it; the policy itself does no I/O.

class `RetryPolicy` DATACLASS

`frozen=True` .

FIELD	TYPE	DEFAULT	DESCRIPTION
<code>max_attempts</code>	<code>int</code>	<code>3</code>	Total attempts including the first try. Set to 1 to disable retries.
<code>base_delay</code>	<code>float</code>	<code>0.2</code>	Seconds to wait before the second attempt. Each subsequent attempt doubles up to <code>max_delay</code> .
<code>max_delay</code>	<code>float</code>	<code>30.0</code>	Cap on the computed delay before jitter.
<code>jitter</code>	<code>float</code>	<code>0.1</code>	Fraction of the delay to randomize by, in <code>[0, 1]</code> . With <code>jitter=0.1</code> and a 1-second delay, the actual sleep is uniformly in <code>[0.9, 1.1]</code> .

```
def delay_for(attempt: int, retry_after: float | None = None) → float
```

Compute the sleep duration before `attempt` (1-indexed). `attempt=1` returns 0 (no delay before the first try).

PARAMETER	TYPE	DEFAULT	DESCRIPTION
-----------	------	---------	-------------

<code>attempt</code>	<code>int</code>	—	The upcoming attempt number, starting at 2.
<code>retry_after</code>	<code>float None</code>	<code>None</code>	Optional server-supplied hint (from a <code>Retry-After</code> header). When set, returned verbatim; jitter is not applied (the server's hint is authoritative).

RETRYABLE_STATUS_CODES

Module constant: `frozenset({429, 502, 503, 504})` . Intentionally not configurable; if your platform deployment legitimately returns transient 500 s, fix the platform.

Example

```
from kneo_client import KneoClient
from kneo_client.core.retries import RetryPolicy

# Aggressive policy for a flaky network
client = KneoClient(profile, retry_policy=RetryPolicy(max_attempts=8,
base_delay=0.5))

# No retries (first failure surfaces immediately)
client = KneoClient(profile, retry_policy=RetryPolicy(max_attempts=1))
```

Idempotency helpers

Every `POST` the client sends carries an `Idempotency-Key` . `Transport` auto-injects a fresh UUID4 per request unless the caller supplies one via the method's `idempotency_key=` parameter.

The platform short-circuits a duplicate `POST` with the same key + identical payload, returning the original response. Reuse of the same key with a different payload yields HTTP 409, surfaced as

[`KneoIdempotencyMismatchError`](#) .

CONSTANT	VALUE	PURPOSE
IDEMPOTENCY_KEY_HEADER	"Idempotency-Key"	The header name as the platform expects it.
MAX_KEY_LENGTH	256	Platform-enforced maximum length for the value.

```
def new_idempotency_key() → str
```

Return a fresh UUID4-based idempotency key as a string.

```
def validate_idempotency_key(key: str) → None
```

Validate an externally-supplied key. Raises `ValueError` if empty or longer than `MAX_KEY_LENGTH`.

List-method results

Module `kneo_client.core.results` (renamed from `core.pagination` in 0.2.0). Hosts the two collection wrappers every list-method on `PlatformClient` returns, plus the page-walking `iterate_all()`.

class Page[T] DATACLASS

Wrapper around list-shaped server responses. Metadata fields are all Optional — an endpoint that doesn't echo a field (notably `audit.list`, which is `limit`-only) leaves that field as `None` rather than carrying a synthesized value. Supports sequence ergonomics: `__iter__`, `__len__`, `__getitem__`.

FIELD	TYPE	DESCRIPTION
items	<code>list[T]</code>	Items on this page.
total	<code>int None</code>	Total items across all pages, when the server echoes it.
limit	<code>int None</code>	Page size the server applied, when echoed.
offset	<code>int None</code>	Offset of the first item on this page, when echoed.

<code>sort_by</code>	<code>str None</code>	Sort field in effect, when echoed.
<code>sort_order</code>	<code>str None</code>	"asc" or "desc" , when echoed.

Properties

PROPERTY	TYPE	DESCRIPTION
<code>count</code>	<code>int</code>	<code>len(items)</code> .
<code>has_more</code>	<code>bool</code>	<code>offset + count < total</code> ; False when <code>offset</code> or <code>total</code> is <code>None</code> (server didn't tell us, so we don't speculate).

class `Map[K, V]` DATACLASS

Wrapper around dict-keyed server responses (`credentials.list` , `policies.environment_list`). The whole collection comes back in one call; there is no pagination. Exposes dict-style accessors.

FIELD	TYPE	DESCRIPTION
<code>items</code>	<code>Mapping[K, V]</code>	The underlying keyed collection.

Properties + methods

MEMBER	TYPE	DESCRIPTION
<code>count</code>	<code>int</code>	<code>len(items)</code> .
<code>__getitem__</code>	<code>(K) → V</code>	Bracket lookup.
<code>__iter__</code>	<code>→ Iterator[V]</code>	Iterates <i>values</i> (consistent with <code>Page</code> iterating items).
<code>__contains__</code>	<code>(object) → bool</code>	Key-based membership.
<code>get</code>	<code>(K, default=None) → V None</code>	Safe lookup.
<code>keys</code>	<code>() → KeysView[K]</code>	Keys view.
<code>values</code>	<code>() → ValuesView[V]</code>	Values view.

```
async def iterate_all(fetch_page: Callable[[int, int], Awaitable[Page[T]]], *,
page_size: int = 100, start_offset: int = 0) → AsyncIterator[T]
```

Walk all items across pages. `fetch_page(limit, offset)` must return a `Page`. Clamps `page_size` to `MAX_PAGE_SIZE = 1000`. Stops when `Page.has_more` is `False` — either because the server signaled end-of-data or because metadata fields are `None` (audit endpoint case; one fetch then exit).

Constants: `DEFAULT_PAGE_SIZE = 100`, `MAX_PAGE_SIZE = 1000`.

Request-ID helpers

Every request carries an `X-Request-ID`. The transport auto-injects a UUID4; callers can override via the method's `request_id=` parameter for cross-service correlation. The platform echoes the ID on responses and in audit events.

CONSTANT	VALUE
<code>REQUEST_ID_HEADER</code>	<code>"X-Request-ID"</code>

```
def generate_request_id() → str
```

Return a fresh UUID4-based request ID as a string.

Logging helpers

Loggers under the `kneo_client.*` namespace use the standard logging machinery. All header payloads are passed through `redact_headers()` before any sink sees them.

CONSTANT	VALUE
<code>REDACTED</code>	<code>"<redacted>"</code>

```
def get_logger(name: str = "kneo_client") → logging.Logger
```

Return a logger under the `kneo_client.*` hierarchy. Bare names like `"transport"` are namespaced to `kneo_client.transport`; already-namespaced names (e.g. `"kneo_client.foo"`) pass through

unchanged.

```
def redact_headers(headers: Mapping[str, str]) → dict[str, str]
```

Return a copy of `headers` with sensitive values replaced by `<redacted>`. Sensitive header tokens (case-insensitive, substring match): `authorization`, `x-kneo-api-key`, `cookie`, `set-cookie`, `proxy-authorization`.

Body redaction is the caller's responsibility. The redaction layer masks headers, not bodies. If you log request / response payloads, filter them at your application's logging-formatter layer.

Exceptions

Every failure surfaces as a typed exception derived from `KneoError`. Each one carries the original HTTP status, response body (parsed JSON when possible), the server-assigned request ID, and — for `POST` failures — the idempotency key that was sent.

Hierarchy

```
KneoError |— KneoNetworkError # transport-level: DNS / connect / TLS / read
timeout |— KneoAuthError # HTTP 401 |— KneoPermissionError # HTTP 403 |—
KneoNotFoundError # HTTP 404 |— KneoConflictError # HTTP 409 (generic) |—
KneoIdempotencyMismatchError # HTTP 409 on idempotency-key replay with different
payload |— KneoRateLimited # HTTP 429 (carries .retry_after) |—
KneoServerError # HTTP 5xx |— KneoServiceUnavailable # HTTP 503 – backpressure
/ overload (carries .retry_after)
```

class KneoError EXCEPTION

Base exception. Inherits from `Exception`.

ATTRIBUTE	TYPE	DESCRIPTION
<code>status</code>	<code>int</code> <code>None</code>	HTTP status, or <code>None</code> for transport-level failures.

body	Any	Parsed JSON dict, raw text, or None .
request_id	str None	The X-Request-ID the platform returned, when present.
idempotency_key	str None	The Idempotency-Key sent on the failing request, when set.

class KneoRateLimited EXCEPTION

Adds one attribute on top of KneoError :

ATTRIBUTE	TYPE	DESCRIPTION
retry_after	float None	Seconds parsed from the Retry-After header.

class KneoServiceUnavailable EXCEPTION

HTTP 503 — the platform is temporarily unavailable or shedding load (notably kneo_serv 's run-queue backpressure on POST /v1/runs). A **subclass of KneoServerError** , so except KneoServerError still catches it. The transport already auto-retries 503 (honoring Retry-After); this surfaces only once retries are exhausted. Adds one attribute on top of KneoError :

ATTRIBUTE	TYPE	DESCRIPTION
retry_after	float None	Seconds parsed from the Retry-After header, or None when the server did not provide the hint.

from_response() FUNCTION

```
def from_response(response: httpx.Response, *, idempotency_key: str | None = None) → KneoError
```

Map an HTTP response to the appropriate KneoError subclass.

HTTP status	Exception	Notes
401	KneoAuthError	Missing or invalid API key.
403	KneoPermissionError	Key valid; insufficient scope.
404	KneoNotFoundError	Resource missing.
409 (with key)	KneoIdempotencyMismatchError	Idempotency-key replay with different payload.
409 (no key)	KneoConflictError	Generic conflict.
429	KneoRateLimited	Carries <code>retry_after</code> .
503	KneoServiceUnavailable	Backpressure / overload. KneoServerError subclass; carries <code>retry_after</code> .
5xx (other)	KneoServerError	Server-side failure.
Other	KneoError	Catch-all.
Transport	KneoNetworkError	Wrapped from <code>httpx.HTTPError</code> .

Example

```
from kneo_client.core.errors import KneoError, KneoIdempotencyMismatchError,
KneoRateLimited

try:
    run = await client.platform.runs.create(payload, idempotency_key=key)
except KneoIdempotencyMismatchError as exc:
    # Same key reused with a different payload – almost always a caller bug.
    log.error("mismatch on key=%r body=%r", exc.idempotency_key, exc.body)
    raise
except KneoRateLimited as exc:
    await asyncio.sleep(exc.retry_after or 10)
except KneoError as exc:
    log.error("create_run failed status=%s rid=%s", exc.status, exc.request_id)
    raise
```

PlatformClient

Operational surface for **Kneo Agent Dashboard**. Aggregates six sub-clients backed by a shared [Transport](#).

class PlatformClient CLASS

ATTRIBUTE	TYPE	DESCRIPTION
health	HealthClient	/v1/{healthz,livez,readyz }
runs	RunsClient	/v1/runs + 11 sub-endpoints
human_tasks	HumanTasksClient	/v1/human-tasks family
audit	AuditClient	/v1/audit-events
credentials	CredentialsClient	/v1/security/credentials
policies	PoliciesClient	/v1/policies/environment family


```
def __init__(transport: Transport) → None
```

Build the aggregator and attach all six sub-clients. Usually you don't construct one directly — `KneoClient(profile).platform` gives you one.

HealthClient

Wraps the three platform health probes.

class HealthClient CLASS

```
async def healthz() → HealthResponse
```

GET `/v1/healthz` — overall service health.

```
async def livez() → HealthResponse
```

GET `/v1/livez` — process liveness. Does *not* check downstream deps.

```
async def readyz() → HealthResponse
```

GET `/v1/readyz` — readiness (database, queue, runtime registry, providers, MCP).

Response (HealthResponse)

FIELD	TYPE	DESCRIPTION
<code>ok</code>	<code>bool</code>	Overall pass/fail.
<code>service</code>	<code>str</code> <code>UNSET</code>	Service name, e.g. "kneo-serv-platform" .
<code>version</code>	<code>str</code> <code>UNSET</code>	Platform version.
<code>metadata</code>	<code>HealthResponseMetadata</code> <code>UNSET</code>	Per-check breakdown.

RunsClient

The largest sub-client — 12 endpoints + one convenience helper (`wait_for_completion`). Covers the full run lifecycle: create, list, get, cancel, continue, replay, trace, checkpoints, policy reports, recovery.

class RunsClient CLASS

```
async def create(body: RunCreateRequest | dict, *, idempotency_key: str | None = None) → RunCreateResponse
```

POST /v1/runs — start a new run. Idempotency key is auto-generated unless caller supplies one.

```
async def list(*, status: str | None = None, limit: int = 100, offset: int = 0, sort_by: str = "updated_at", sort_order: str = "desc") → Page[RunListResponseRunsItem]
```

GET /v1/runs — list runs with limit/offset pagination.

```
async def get(run_id: str) → RunStatusResponse
```

GET /v1/runs/{run_id} — fetch the current status of a run.

```
async def cancel(run_id: str, *, idempotency_key: str | None = None) → RunStatusResponse
```

POST /v1/runs/{run_id}/cancel — cooperatively cancel a running run. The platform may not honor the cancel if the run is already in a terminal state; the returned status reflects the post-cancel state.

```
async def continue_(run_id: str, *, idempotency_key: str | None = None) → RunCreateResponse
```

POST /v1/runs/{run_id}/continue — resume a paused run (typically after a human task was resumed). Trailing underscore avoids the Python `continue` keyword collision.

```
async def replay(run_id: str) → RunReplayResponse
```

GET /v1/runs/{run_id}/replay — fetch a deterministic replay view derived from the run's checkpoints + trace.

```
async def trace(run_id: str, *, event_type: str | None = None, limit: int = 100, offset: int = 0, sort_by: str = "timestamp", sort_order: str = "asc") → Page[TraceResponseEventsItem]
```

GET /v1/runs/{run_id}/trace — fetch the event trace for a run (tool calls, model calls, middleware decisions, etc.).

```
async def checkpoints(run_id: str, *, type: str | None = None, limit: int = 100, offset: int = 0, sort_by: str = "sequence", sort_order: str = "asc") → Page[CheckpointListResponseCheckpointsItem]
```

GET /v1/runs/{run_id}/checkpoints — list a run's serialized state snapshots.

```
async def checkpoints_diff(run_id: str, *, from_sequence: int | None = None, to_sequence: int | None = None) → CheckpointDiffResponse
```

GET /v1/runs/{run_id}/checkpoints/diff — diff two checkpoints.

```
async def policy_report(run_id: str) → dict[str, Any]
```

GET /v1/runs/{run_id}/policy-report — fetch policy outcomes for a run. Returns the raw JSON body; the platform's policy-report schema is intentionally open-ended and not modeled as a typed response.

```
async def recovery(run_id: str) → RunRecoveryResponse
```

GET /v1/runs/{run_id}/recovery — fetch the recovery context for a failed run (what state can be recovered, what next action is recommended).

```
async def wait_for_completion(run_id: str, *, timeout: float | None = None, poll_interval: float = 1.0, terminal_statuses: Iterable[str] | None = None) → RunStatusResponse
```

Poll `get(run_id)` until the run reaches a terminal status. Convenience helper; not a separate platform endpoint.

PARAMETER	TYPE	DEFAULT	DESCRIPTION
<code>timeout</code>	<code>float None</code>	<code>None</code>	Total time budget in seconds. <code>None</code> means wait indefinitely.

<code>poll_interval</code>	<code>float</code>	<code>1.0</code>	Seconds to sleep between polls.
<code>terminal_statuses</code>	<code>Iterable[str] None</code>	<code>None</code>	Defaults to <code>{"completed", "failed", "cancelled"}</code> . Pass <code>{"paused_human_review", ...}</code> to treat additional states as terminal.

RETURNS

The terminal `RunStatusResponse` .

RAISES

`TimeoutError` if `timeout` elapses before a terminal status is reached.

Module constant: `DEFAULT_TERMINAL_STATUSES = frozenset({"completed", "failed", "cancelled"})` .

```
async def tail_trace(run_id: str, *, poll_interval: float = 1.0, timeout: float | None = None, terminal_statuses: Iterable[str] | None = None, start_offset: int = 0, page_size: int = 100) → AsyncIterator[TraceResponseEventsItem]
```

Stream trace events as they arrive. Polls `trace(run_id)` with ascending offset, yields each new event as the page arrives, then performs one final drain pass after the run reaches a terminal status to capture any events emitted between the last poll and the status transition. Returns when the drain pass finds no new events. Convenience helper; not a separate platform endpoint.

PARAMETER	TYPE	DEFAULT	DESCRIPTION
<code>poll_interval</code>	<code>float</code>	<code>1.0</code>	Seconds to sleep between polls.
<code>timeout</code>	<code>float None</code>	<code>None</code>	Total time budget. <code>None</code> means wait indefinitely.
<code>terminal_statuses</code>	<code>Iterable[str] None</code>	<code>None</code>	Status set that ends streaming. Defaults to

```
    {"completed",  
     "failed",  
     "cancelled"} .
```

<code>start_offset</code>	<code>int</code>	<code>0</code>	Offset to start from. Use a saved checkpoint to resume mid-trace.
<code>page_size</code>	<code>int</code>	<code>100</code>	Per-poll page size; clamped server-side to 1000.

RAISES

`TimeoutError` if `timeout` elapses before a terminal status is reached.

When to pick this over `wait_for_completion` — see [polling and waiting guide](#). Short version:

`wait_for_completion` when you only need the final status; `tail_trace` when you need to display, log, or react to events as they arrive (typically interactive UIs and dev tooling).

Example: end-to-end run with live trace

```
async with KneoClient.from_profile() as client:  
    created = await client.platform.runs.create({"spec_id": "my-spec"})  
    async for ev in client.platform.runs.tail_trace(created.run_id, timeout=600):  
        print(f"{ev['sequence']}: {ev['event_type']}")  
    # tail_trace returned → the run reached a terminal status and the drain pass  
    # found no more events.  
    terminal = await client.platform.runs.get(created.run_id)  
    print(f"final={terminal.status}")
```

HumanTasksClient

Human-in-the-loop pause points. A run that requires operator review surfaces as a *human task*, identified by a `continuation_id`. Resuming posts a decision and unblocks the paused continuation.

class HumanTasksClient CLASS

```

async def list(*, status: str | None = None, run_id: str | None = None,
workflow_kind: str | None = None, limit: int = 100, offset: int = 0, sort_by:
str = "created_at", sort_order: str = "desc") →
Page[HumanTaskListResponseTasksItem]

```

GET /v1/human-tasks — list pending and recent human tasks as a `Page`. `run_id` and `workflow_kind` are server-side filters (added in 0.2.0). `status` is honored by the server but is a wrapper-side kwarg per a documented spec-side omission (see [TODO-0.3.0 § parameter-drift gate](#)).

```

async def get(continuation_id: str) → HumanTaskResponse

```

GET /v1/human-tasks/{continuation_id} — fetch a single human task.

```

async def resume(continuation_id: str, body: HumanResumeRequest | dict, *,
idempotency_key: str | None = None) → HumanResumeResponse

```

POST /v1/human-tasks/{continuation_id}/resume — resume a paused run with a decision.

```

async def resume_by_id(continuation_id: str, body: HumanResumeRequest | dict,
*, idempotency_key: str | None = None) → HumanResumeResponse

```

Fetch a task by ID, then resume it. Calls `get(continuation_id)` first so a stale `continuation_id` surfaces as `KneoNotFoundError` from the GET rather than a conflict from the POST. Use this when distinguishing "the task is gone" from "the resume payload conflicts" matters; call `resume()` directly otherwise.

```

async def resume_first_pending(body: HumanResumeRequest | dict, *, run_id: str
| None = None, workflow_kind: str | None = None, idempotency_key: str | None =
None) → HumanResumeResponse | None

```

List pending tasks matching the filters and resume the first match; returns `None` if no matching pending task exists. Worker-pattern convenience: collapses the list-then-resume dance into one call. The list call uses `limit=1` and `status="pending"`.

PARAMETER	TYPE	DEFAULT	DESCRIPTION
<code>run_id</code>	<code>str</code> <code>None</code>	<code>None</code>	Filter — only consider tasks for this run.
<code>workflow_kind</code>	<code>str</code> <code>None</code>	<code>None</code>	Filter — only consider tasks of this workflow

kind.

<code>idempotency_key</code>	<code>str None</code>	<code>None</code>	Optional caller-supplied key for the POST ; transport generates a fresh UUID4 by default.
------------------------------	-------------------------	-------------------	---

AuditClient

Audit events are the canonical operational log: every run, human-task, credential, and policy mutation produces one.

class AuditClient CLASS

```
async def list(*, event_type: str | None = None, run_id: str | None = None,
limit: int = 100) → Page[AuditEventListResponseEventsItem]
```

GET /v1/audit-events — list audit events as a Page . The endpoint is limit -only: it accepts solely event_type / run_id / limit server-side, and the Page returned has offset / total / sort_* as None (server-side limit-only, honestly surfaced rather than synthesized). *(The principal / offset / sort_by / sort_order kwargs — deprecated in 0.4.0, never honored by the platform — were removed in 0.5.0.)*

CredentialsClient

Lists the credential *references* the platform knows about. The platform never returns raw secret material via the HTTP API.

class CredentialsClient CLASS

```
async def list(*, providers: list[str] | None = None, extras: list[str] | None
= None, include_service_tokens: bool = True) → Map[str, Any]
```

GET `/v1/security/credentials` — list known credential references as a `Map` keyed by credential ID. Credential body is open-shaped in the pinned spec, so values are `Any`. The whole inventory is returned in one call (no pagination). Filter via `providers=["aws", "gcp"]`, `extras=["scope"]`, or `include_service_tokens=False`; the three kwargs were added in 0.3.0.

PoliciesClient

Environment policies map deployment targets (e.g. `dev`, `staging`, `prod`) to policy bundles that gate which specs / agents are allowed to run there.

class PoliciesClient CLASS

```
async def environment_list() → Map[str,
EnvironmentPolicyListResponsePoliciesAdditionalProperty]
```

GET `/v1/policies/environment` — list policies for every environment as a `Map` keyed by environment name. The whole inventory is returned in one call (no pagination).

```
async def environment_get(environment: str) → EnvironmentPolicyResponse
```

GET `/v1/policies/environment/{environment}` — fetch one environment's policy.

```
async def environment_put(environment: str, body: EnvironmentPolicyRequest |
dict) → EnvironmentPolicyResponse
```

PUT `/v1/policies/environment/{environment}` — replace an environment's policy.

Note. PUT is idempotent by HTTP semantics; the transport does not inject an `Idempotency-Key` for it.

AgentClient

Development surface for **Kneo Agent Studio**. Currently aggregates one sub-client ([SpecsClient](#)).

class AgentClient CLASS

ATTRIBUTE	TYPE	DESCRIPTION
<code>specs</code>	SpecsClient	<code>/v1/specs/*</code>

SpecsClient

Spec validation, compilation, explanation, policy-report, and ad-hoc dry-run. The Studio iterate-and-test loop.

class SpecsClient CLASS

```
async def validate(body: SpecValidateRequest | dict, *, idempotency_key: str | None = None) → SpecValidateResponse
```

POST `/v1/specs/validate` — schema + semantic validation of a spec.

```
async def compile(body: SpecCompileRequest | dict, *, idempotency_key: str | None = None) → SpecCompileResponse
```

POST `/v1/specs/compile` — compile a spec to its runtime representation.

```
async def explain(body: SpecExplainRequest | dict, *, idempotency_key: str | None = None) → SpecExplainResponse
```

POST `/v1/specs/explain` — human-readable summary of a spec.

```
async def policy_report(body: SpecPolicyReportRequest | dict, *, idempotency_key: str | None = None) → SpecPolicyReportResponse
```

POST `/v1/specs/policy-report` — preview policy outcomes for a spec.

```
async def run(body: RunCreateRequest | dict, *, idempotency_key: str | None = None) → RunCreateResponse
```

POST `/v1/specs/run` — ad-hoc dry-run of a spec. Distinct from [RunsClient.create](#): `specs.run` accepts an inline spec rather than a spec reference, and is intended for Studio's iterate-and-test flow.

```

async def validate_then_compile(body: SpecValidateRequest | dict, *,
    idempotency_key: str | None = None) → ValidateThenCompileResult

```

Validate first, then compile only if validate succeeded. Skips the compile call on validate failure to avoid a wasted round-trip on a known-bad spec; `result.compile` is `None` in that case and `result.ok` returns `False`. Same `body` shape as `validate / compile`; the helper passes the same payload to both calls.

```

async def dry_run(body: SpecValidateRequest | dict, *, idempotency_key: str |
    None = None) → DryRunResult

```

Validate, then run `explain` + `policy_report` in parallel via `asyncio.gather` once validate succeeds. Skips both downstream calls on validate failure. Returns a frozen `DryRunResult` with the validate result, the (optional) `explain` and `policy-report` responses, and an `.ok` property.

Example: validate-then-compile via the helper

```

async with KneoClient.from_profile() as client:
    payload = {"spec": spec_text}
    result = await client.agent.specs.validate_then_compile(payload)
    if not result.ok:
        for d in (result.validate.diagnostics or []):
            print(f"validate: {d}")
        if result.compile is not None:
            for d in (result.compile.diagnostics or []):
                print(f"compile: {d}")
    return
    # result.compile is the compiled spec; ship it.

```

ValidateThenCompileResult

Frozen dataclass returned by `SpecsClient.validate_then_compile`. Bundles the validate and (optional) compile responses with a single boolean for the common "is the spec shippable?" question.

class ValidateThenCompileResult DATACLASS

FIELD	TYPE	DESCRIPTION
<code>validate</code>	<code>SpecValidateResponse</code>	Always populated.
<code>compile</code>	<code>SpecCompileResponse</code> <code>None</code>	<code>None</code> when <code>validate</code> failed (the helper skipped the compile call).

Properties

PROPERTY	TYPE	DESCRIPTION
<code>ok</code>	<code>bool</code>	True iff <code>validate.valid</code> AND <code>compile</code> is not <code>None</code> AND <code>compile.ok</code> .

DryRunResult

Frozen dataclass returned by `SpecsClient.dry_run`. Bundles `validate` + `explain` + `policy-report` into a single object for Studio's pre-ship dashboard.

class DryRunResult DATACLASS

FIELD	TYPE	DESCRIPTION
<code>validate</code>	<code>SpecValidateResponse</code>	Always populated.
<code>explain</code>	<code>SpecExplainResponse</code> <code>None</code>	<code>None</code> when <code>validate</code> failed.
<code>policy_report</code>	<code>SpecPolicyReportResponse</code> <code>None</code>	<code>None</code> when <code>validate</code> failed.

Properties

PROPERTY	TYPE	DESCRIPTION
<code>ok</code>	<code>bool</code>	True iff <code>validate</code> succeeded AND <code>policy_report</code> is not <code>None</code> AND <code>policy_report.valid</code> is not <code>False</code> . The <code>UNSET</code> case for

`policy_report.valid` is treated as "no objection" (returns `True`) — absence is not failure.

Compatibility matrix

Which `kneo-client` release supports which `kneo_serv` platform version. The platform's `/v1` HTTP API is a stability boundary.

kneo-client	Pinned to kneo_serv	Tested against	Python	Status
0.5.0	v0.5.0 (<code>info.version</code> 0.5.0)	kneo_serv 0.5.x line	≥ 3.12 (3.12 / 3.13 / 3.14 in CI)	Current
0.4.0	v0.4.0 (<code>info.version</code> 0.4.0)	kneo_serv 0.4.x line	≥ 3.12 (3.12 / 3.13 / 3.14 in CI)	Previous
0.3.0	v0.4.0 (<code>info.version</code> 0.4.0)	kneo_serv 0.4.x line	≥ 3.12 (3.12 / 3.13 / 3.14 in CI)	Older

Forward compatibility

A newer `kneo_serv` that adds endpoints will still work for everything `kneo-client` already wraps. New endpoints are available via the drop-to-transport escape hatch until the next `kneo-client` release wraps them:

```
async with KneoClient.from_profile() as client:
    resp = await client._transport.request("GET", "/v1/some/new/endpoint")
    payload = resp.json()
```

Backward compatibility

kneo-client X.Y.Z is **not** guaranteed against kneo_serv releases older than its pin. Wrappers may rely on response fields that older versions don't emit, and error mapping assumes the current platform error shape. Pin to a matching kneo-client minor when talking to an older platform.

Guides

Walking pages with `iterate_all()`

List methods return `Page` (or `Map` for the two dict-keyed endpoints) directly — no adapter needed. Pass the list method itself as the `fetch_page` callable:

```
from kneo_client.core.results import iterate_all

async def all_runs(client, **filters):
    async def fetch_page(limit, offset):
        return await client.platform.runs.list(limit=limit, offset=offset,
        **filters)
    async for run in iterate_all(fetch_page, page_size=200):
        yield run
```

For the audit endpoint, `iterate_all` does one fetch and exits because `has_more` is `False` (server returns only `count + events`; no offset to advance). For the two `Map`-returning endpoints, walk the values directly: `for cred in (await client.platform.credentials.list()): .`

Bring your own httpx client

For shared connection pools, custom transports, proxy configuration:

```

import httpx
from kneo_client.core.auth import ApiKeyAuth, AuthScheme
from kneo_client.core.profiles import Profile
from kneo_client.core.transport import Transport

profile = Profile(name="prod", url="https://kneo", api_key="...",
                  auth_scheme=AuthScheme.BEARER, timeout=30.0)
shared = httpx.AsyncClient(
    base_url=profile.url,
    auth=ApiKeyAuth(profile.api_key, profile.auth_scheme),
    timeout=profile.timeout,
    limits=httpx.Limits(max_keepalive_connections=20, max_connections=100),
)
try:
    async with Transport(profile, http_client=shared) as t:
        # t.aclose() will NOT close `shared`.
        ...
finally:
    await shared.aclose()

```

Wait for a non-default terminal status

`wait_for_completion()` treats {"completed", "failed", "cancelled"} as terminal by default. To return as soon as the run pauses for human review:

```

terminal = await client.platform.runs.wait_for_completion(
    run_id,
    poll_interval=2.0,
    timeout=300,
    terminal_statuses={"completed", "failed", "cancelled", "paused_human_review"},
)

```

More recipes

See `docs/dev/extending.md` on the repo for the full 10-recipe set: custom retry policy, custom auth scheme, wrapping a not-yet-adapter-covered endpoint, adding a new wrapped endpoint, custom logging,

profile from a secrets manager, paginated iteration, custom terminal statuses, sync facade, and what NOT to extend.

Kneo Agent Client — API Reference Manual · v0.5.0 · 2026-05-29 · MIT license

Adapted layout and style from the `kneo_agent` SDK API reference.